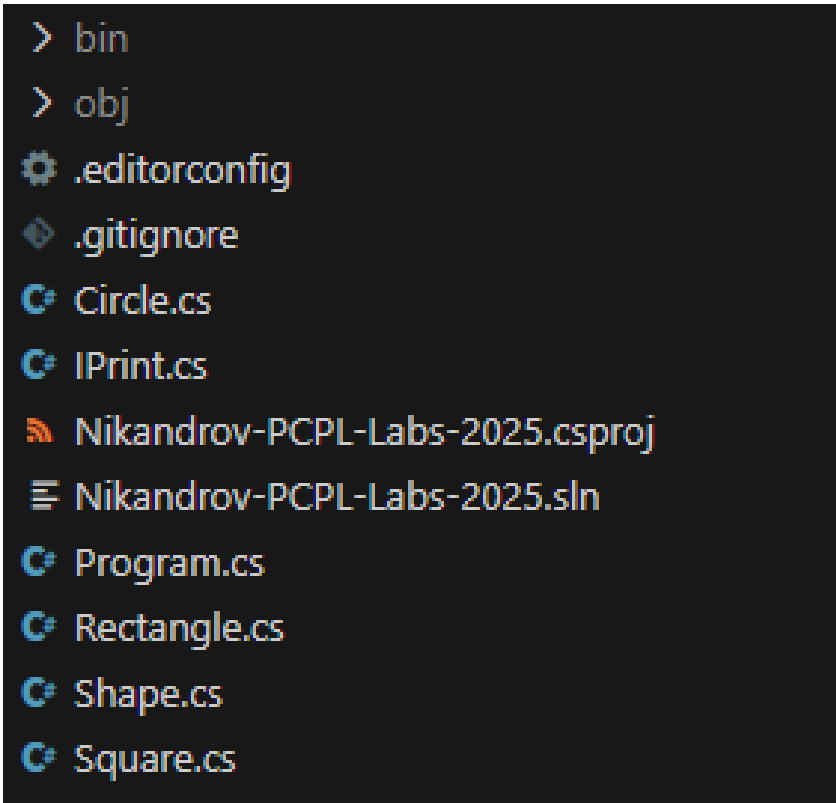


Отчет по лабораторной работе №2 по дисциплине ПиКЯП

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Файловая структура проекта:



```
> bin
> obj
⚙ .editorconfig
🔒 .gitignore
🔗 Circle.cs
🔗 IPrint.cs
🔗 Nikandrov-PCPL-Labs-2025.csproj
📄 Nikandrov-PCPL-Labs-2025.sln
🔗 Program.cs
🔗 Rectangle.cs
🔗 Shape.cs
🔗 Square.cs
```

Листинг кода проекта:

```
using Shapes;

class Program
{
    static void Main(string[] args)
    {
        Rectangle rectangle = new(2.0, 3.0);
        Square square = new(2);
        Circle circle = new(3);

        rectangle.Print();
        square.Print();
        circle.Print();
    }
}
```

```
public interface IPrint
{
    public void Print();
}
```

```
namespace Shapes;

public abstract class Shape
{
    public virtual double CalculateArea() => 0.0;
}
```

```
namespace Shapes;

public class Rectangle : Shape, IPrint
{
    public Rectangle(double width, double height)
    {
        Width = Math.Abs(width);
        Height = Math.Abs(height);
    }

    public double Width { get; init; }
    public double Height { get; init; }

    public void Print() => Console.WriteLine(ToString());
}
```

```
    public override double CalculateArea() => Width * Height;

    public override string ToString() => $"Width: {Width:F4}; Height:
{Height:F4}; Area: {CalculateArea():F4}";
}
```

```
namespace Shapes;

public class Square : Rectangle
{
    public Square(double length) : base(length, length) { }

    public override string ToString() => $"Side length: {Width:F4}; Area:
{CalculateArea():F4}";
}
```

```
namespace Shapes;

public class Circle : Shape, IPrint
{
    public Circle(double radius)
    {
        Radius = Math.Abs(radius);
    }

    public double Radius { get; init; }

    public void Print() => Console.WriteLine(ToString());

    public override double CalculateArea() => Math.PI * Radius * Radius;

    public override string ToString() => $"Radius: {Radius:F4}; Area:
{CalculateArea():F4}";
}
```

Результат работы программы:

```
vIntester@DESKTOP-ACISLUE:~/Nikandrov-PCPL-Labs-2025$ dotnet run
Width: 2.0000; Height: 3.0000; Area: 6.0000;
Side length: 2.0000; Area: 4.0000;
Radius: 3.0000; Area: 28.2743
```